

Neil Malur

Cambridge, MA | malur@mit.edu | +1-339-222-4916 | neilmalur.com | linkedin.com/in/nmalur

Education

Massachusetts Institute of Technology, BS in Computer Science and Engineering Sept 2021 – May 2025

- GPA: 5.0/5.0 | IEEE Eta Kappa Nu Honor Society | Social and Ethical Responsibilities of Computing Scholar
- **Coursework:** Computer Systems Security, Computer Networks, Natural Language Processing, Software Construction
- **Activities:** South Asian Association of Students (President), Bhangra (Captain), DynaMIT Mentor, Asian American Initiative (Executive Board), AI@MIT Labs, Mock Trial, Student Government

Experience

Incoming Software Engineer, Meta – New York, NY Starting Sep 2025

Incoming Research Intern, Indian Institute of Science – Bengaluru, Karnataka Starting Jun 2025

Software Engineer Intern, Nominal – New York, NY May 2024 – Aug 2024

- Streamlined file ingest by implementing timeseries database read and caching from AWS S3, cutting upload time by 93%. Removed writes to Apache Arrow, instead directing CSV upload to ClickHouse from S3 before caching the timeseries.
- Developed and designed ability to generate end-to-end HTTP API connections for user data, built weather data instances for customers. Connections loaded in live and historical data into plotting service optimized for customer ease of use.
- Investigated thread control changes, including semaphore-enforced limits by user organization and migration to Temporal
- Augmented compute node tree and resolution to support parameterized peak-finding and inverse trigonometric functions

Summer Analyst Intern, United States Department of Commerce – Washington, DC May 2023 – Aug 2023

- Developed model with 7-15 generated macroeconomic zones alongside reusable mapping software for grant deliberation
- Authored economic study of the drivers of Charlotte's transformation with stakeholder interviews and original analysis
- Spearheaded creation of a machine learning and software pipeline at Commerce, prepared research for Under Secretary and House, and refined bureau strategy

Student Researcher, MIT Climate and Sustainability Consortium – Cambridge, MA May 2022 – Jan 2023

- Created graph learning piece and training strategy of hybrid reinforcement learning system to predict polymer properties.
- Built software interface to simulate crystallization with molecular dynamics, allowing chemists to interact with model
- Coauthored paper presented at ICML Workshop on Generative Modeling

Software Engineer Intern, Conjuguemos – Remote June 2020 – Aug 2021

- Launched entire-classroom registration for schools during COVID-19 with OAuth 2.0 third-party SSO with Google
- Restructured database to add assignments tied to specific classes and automated third-party assignment integration

Projects

Security Key Login

- Secured login for a social media website under the WebAuthn 2.0 framework to authenticate second-factor security keys
- Obtained web certificates to communicate via HTTPS by collaborating with an ACME-compliant certificate authority

DIE-BERT

- Augmented BERT model to accept data with duration-based token embeddings to estimate the impact of textual prosody
- Evaluated model on masked language objective and by correlating attention values, generated data with forced alignment

Homemade

- Built React-based site with hooks to create recipes based on user-inputted ingredients during intensive competition at MIT
- Deployed website with Heroku, employed MongoDB to store user data, and implemented Google OAuth for login
- Produced interactive and customizable platformer game dynamically generated from API-retrieved recipe information

Technologies

Languages: C++, C, Java, Typescript, Python, Bluespec, React, Node.js, JQuery, SQL, JavaScript

Software: AWS, PyTorch, TensorFlow, Unity, Docker, ClickHouse, Conjure, Datadog, Temporal